

Musculoskeletal and Connective-Tissue Health

Bone, joint, muscle, tendon, spine, movement, pain, rehabilitation,
assistive technology, and measurement engineering

Volume 03I of the Health Atlas Research Suite

Purpose

Connect tissue biology and mechanics to pain, movement, function, participation and device design across injury, chronic disease, disability and aging.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not provide fracture reduction, spine emergency, postoperative, injection or exercise prescriptions. Imaging and wearable outputs must be interpreted against symptoms, examination, function and intended use.

How to use this volume

Musculoskeletal health is not the absence of an imaging abnormality. It is the capacity of bone, joint, muscle, tendon, ligament, connective tissue and neural control to support safe, adaptable movement and valued activity with tolerable symptoms. A complete profile integrates tissue state, load, pain mechanisms, motor strategy, reserve, work/environment, psychosocial context, function and trajectory.

Questions this volume is designed to answer

1. How do tissue capacity, loading, pain and motor control interact?
2. Why do structure, symptoms and disability often disagree?
3. Which measurements capture function and reserve rather than only anatomy?
4. How should wearables, gait systems, prostheses and rehabilitation robots be validated?
5. How can treatment improve participation without medicalizing normal variation or ignoring dangerous disease?

Scope

- Normal bone, joint, muscle, tendon, connective-tissue, movement and repair biology.
- Degenerative, inflammatory, traumatic, infectious, neoplastic, genetic and pain phenotypes.
- Imaging, mechanics, electrophysiology, motion, wearable and tissue measurement.
- Exercise, rehabilitation, pharmacology, assistive technology, procedures and surgery.
- Engineering of biomechanical and human-device systems.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

Contents

Document status and intended use	i
How to use this volume	ii
1 Executive synthesis	1
1.1 What a complete profile requires	1
2 Historical development and changing models	3
2.0.1 Empirical fracture setting, massage, movement practices and descriptions of arthritis predated anatomy.	3
2.0.2 Anatomy, biomechanics, radiography and pathology localized structural lesions and deformity.	4
2.0.3 Orthopedic surgery, antibiotics, joint replacement, rehabilitation professions and standardized imaging transformed care.	4
2.0.4 Immunology, bone biology, pain science, biomaterials and disease-modifying therapy expanded mechanisms.	4
2.0.5 Quantitative imaging, motion sensing, robotics, biologics and participatory rehabilitation connect structure to real-world function.	4
2.0.6 Musculoskeletal medicine is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.	4
3 Normal foundations: structure, physiology, and regulation	5
3.0.1 Bone modeling and remodeling	5
3.0.2 Skeletal muscle excitation, force and adaptation	5
3.0.3 Cartilage, synovium and joint mechanics	6
3.0.4 Tendon, ligament and enthesis biology	6
3.0.5 Connective tissue and fascia	6
3.0.6 Spine and intervertebral system	7
3.0.7 Pain, nociception and protective behavior	7
3.0.8 Movement control, balance and motor learning	7
3.0.9 Repair, regeneration and fibrosis	8
3.0.10 Reserve across the life course	8
3.1 Useful quantitative relationships	8
4 Determinants, heterogeneity, and causal architecture	10
4.0.1 Genetic architecture and family history	12
4.0.2 Development and life course	12
4.0.3 Nutrition and metabolism	12
4.0.4 Physical activity, conditioning, and sedentary behavior	13

4.0.5	Sleep, circadian timing, and recovery	13
4.0.6	Stress, emotion, and behavior	13
4.0.7	Social, economic, cultural, and care conditions	14
4.0.8	Environmental and occupational exposures	14
4.0.9	Health care, medication, and iatrogenic effects	14
5	Major disorders, failure modes, and clinically important phenotypes	16
5.0.1	Osteoarthritis	16
5.0.2	Inflammatory arthritis	16
5.0.3	Osteoporosis and fragility fracture	17
5.0.4	Low back and neck pain	17
5.0.5	Tendinopathy and overuse injury	18
5.0.6	Acute fracture and dislocation	18
5.0.7	Muscle strain, tear and rhabdomyolysis	19
5.0.8	Sarcopenia and frailty-related weakness	19
5.0.9	Inflammatory and inherited myopathy	20
5.0.10	Connective-tissue disorders and hypermobility phenotypes	20
5.0.11	Septic arthritis and osteomyelitis	21
5.0.12	Amputation and limb difference	21
5.0.13	Chronic widespread and nociplastic pain	22
5.0.14	Bone and soft-tissue tumors	22
6	Assessment, clinical reasoning, and longitudinal profiling	24
6.0.1	1. Question and urgency	25
6.0.2	2. History and lived phenotype	25
6.0.3	3. Examination and basic measurements	25
6.0.4	4. Targeted laboratory, imaging, and physiology	26
6.0.5	5. Function and challenge testing	26
6.0.6	6. Longitudinal integration	26
7	Measurement and device landscape	27
7.0.1	History, physical examination and functional task testing	29
7.0.2	Plain radiography and fluoroscopy	30
7.0.3	CT and quantitative CT	30
7.0.4	MRI	30
7.0.5	Musculoskeletal ultrasound	31
7.0.6	DXA and bone-density measurement	31
7.0.7	Force platforms and dynamometry	31
7.0.8	Motion capture and inertial sensing	32
7.0.9	Surface and needle electromyography	32

7.0.10	Gait and plantar-pressure systems	33
7.0.11	Quantitative sensory testing	33
7.0.12	Arthrocentesis and synovial analysis	33
7.0.13	Bone, muscle, synovial or tumor biopsy	34
7.0.14	Wearable activity and load monitoring	34
7.0.15	Elastography and tissue mechanical imaging	34
7.0.16	Smart prosthetic, orthotic and rehabilitation robotics sensing	35
8	Engineering architectures, signal quality, and validation	36
8.0.1	Intended use and measurand	37
8.0.2	Sensor and transduction	37
8.0.3	Timing and synchronization	37
8.0.4	Analog, acquisition, and digital integrity	38
8.0.5	Calibration and traceability	38
8.0.6	Artifact and quality channels	38
8.0.7	Algorithms and uncertainty	39
8.0.8	Safety, usability, cybersecurity, and privacy	39
9	Prevention, treatment, rehabilitation, and system intervention	40
9.0.1	Physical activity, strength and motor training	42
9.0.2	Education, self-management and work/activity modification	42
9.0.3	Analgesic and anti-inflammatory pharmacotherapy	43
9.0.4	Disease-modifying antirheumatic therapy	43
9.0.5	Osteoporosis pharmacotherapy	43
9.0.6	Injections and image-guided procedures	43
9.0.7	Surgery and fixation/reconstruction	44
9.0.8	Orthoses, prostheses and assistive products	44
9.0.9	Fall and injury prevention	44
9.0.10	Psychological and behavioral pain interventions	45
9.0.11	Regenerative, orthobiologic and tissue-engineering interventions	45
9.0.12	Multidisciplinary rehabilitation and vocational reintegration	45
10	What is known, not known, and frequently overstated	46
10.0.1	Can imaging biomarkers predict pain and recovery?	47
10.0.2	Which wearable load metrics prevent injury?	47
10.0.3	How much risk is genetic versus environmental?	48
10.0.4	Can one composite score summarize health?	48
10.0.5	When is earlier detection beneficial?	48
10.0.6	Do digital biomarkers generalize?	49
10.0.7	What counts as meaningful change?	49

11 Research and development roadmap	50
11.0.1 0. Structure-pain-function cohort	51
11.0.2 0. Open rehabilitation sensor stack	51
11.0.3 1. Reference map	52
11.0.4 2. Causal determinants	52
11.0.5 3. Measurement validity	52
11.0.6 4. Clinical utility	52
11.0.7 5. Longitudinal multimorbidity	53
11.0.8 6. Implementation and governance	53
12 Cross-volume integration	54
A Minimum data schema for a longitudinal domain profile	55
B Glossary	56
Selected references and current standards	57

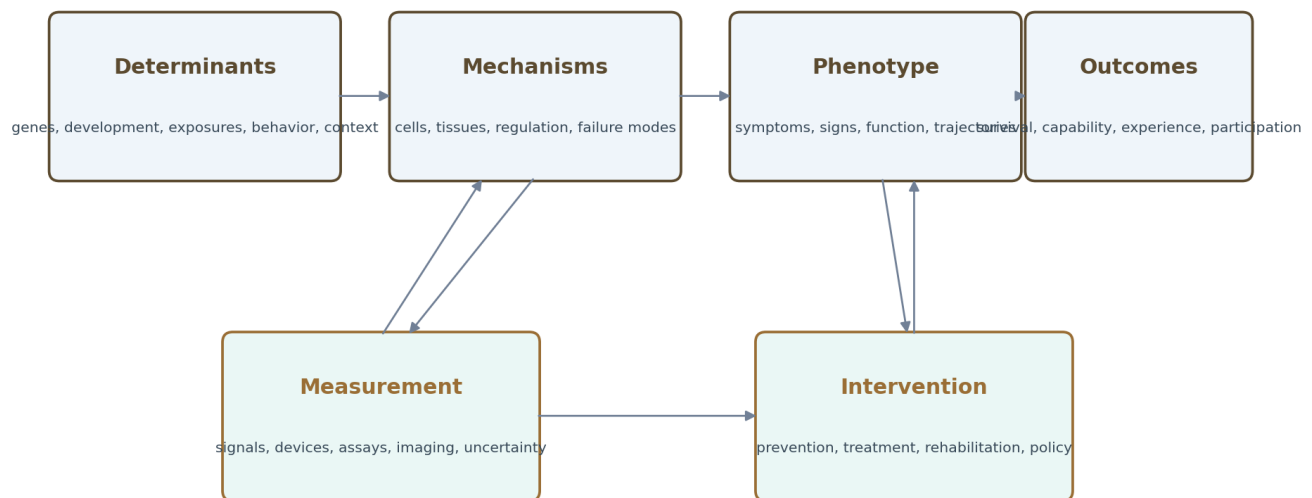
Executive synthesis

Musculoskeletal health is not the absence of an imaging abnormality. It is the capacity of bone, joint, muscle, tendon, ligament, connective tissue and neural control to support safe, adaptable movement and valued activity with tolerable symptoms. A complete profile integrates tissue state, load, pain mechanisms, motor strategy, reserve, work/environment, psychosocial context, function and trajectory.

Central organizing principle

Connect tissue biology and mechanics to pain, movement, function, participation and device design across injury, chronic disease, disability and aging.

Musculoskeletal and Connective-Tissue Health



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Open fracture, compartment syndrome, septic joint, spinal cord/cauda equina signs, major neurovascular compromise, rapidly progressive weakness, malignant bone pain pattern or severe rhabdomyolysis can be emergencies. Human-load devices and powered orthoses require mechanical, electrical, control and usability risk management.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to early modern	Empirical fracture setting, massage, movement practices and descriptions of arthritis predated anatomy.	Established an empirical vocabulary for musculoskeletal medicine while care remained descriptive.	Mechanisms were mixed with humoral, spiritual, and observational explanations.
Seventeenth to nineteenth centuries	Anatomy, biomechanics, radiography and pathology localized structural lesions and deformity.	Connected anatomy and tissue structure to function and disease.	Static specimens could not explain regulation, kinetics, or lived function.
Late nineteenth to mid-twentieth century	Orthopedic surgery, antibiotics, joint replacement, rehabilitation professions and standardized imaging transformed care.	Made quantities, cell populations, and treatment response measurable.	Reference ranges and categories often hid heterogeneity and preanalytic error.
Late twentieth century	Immunology, bone biology, pain science, biomaterials and disease-modifying therapy expanded mechanisms.	Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.	Single-pathway explanations frequently underrepresented networks and context.
Twenty-first century	Quantitative imaging, motion sensing, robotics, biologics and participatory rehabilitation connect structure to real-world function.	Enabled longitudinal, multimodal, high-dimensional, and precision approaches.	Clinical utility, transportability, cost, equity, and data governance remain limiting.
Current systems model	Musculoskeletal medicine is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.	Supports cross-volume reasoning and explicit measurement models.	Education, reimbursement, datasets, and specialties still reinforce silos.

2.0.1 Empirical fracture setting, massage, movement practices and descriptions of arthritis predated anatomy.

Scientific consequence. Established an empirical vocabulary for musculoskeletal medicine while care remained descriptive.

What remained unresolved. Mechanisms were mixed with humoral, spiritual, and observational explanations.

Evidence anchors: [3,13]

2.0.2 Anatomy, biomechanics, radiography and pathology localized structural lesions and deformity.

Scientific consequence. Connected anatomy and tissue structure to function and disease.

What remained unresolved. Static specimens could not explain regulation, kinetics, or lived function.

Evidence anchors: [3,13]

2.0.3 Orthopedic surgery, antibiotics, joint replacement, rehabilitation professions and standardized imaging transformed care.

Scientific consequence. Made quantities, cell populations, and treatment response measurable.

What remained unresolved. Reference ranges and categories often hid heterogeneity and preanalytic error.

Evidence anchors: [3,13]

2.0.4 Immunology, bone biology, pain science, biomaterials and disease-modifying therapy expanded mechanisms.

Scientific consequence. Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.

What remained unresolved. Single-pathway explanations frequently underrepresented networks and context.

Evidence anchors: [3,13]

2.0.5 Quantitative imaging, motion sensing, robotics, biologics and participatory rehabilitation connect structure to real-world function.

Scientific consequence. Enabled longitudinal, multimodal, high-dimensional, and precision approaches.

What remained unresolved. Clinical utility, transportability, cost, equity, and data governance remain limiting.

Evidence anchors: [3,13]

2.0.6 Musculoskeletal medicine is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.

Scientific consequence. Supports cross-volume reasoning and explicit measurement models.

What remained unresolved. Education, reimbursement, datasets, and specialties still reinforce silos.

Evidence anchors: [1,21]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Bone modeling and remodeling

Core principle. Bone adapts through coordinated formation and resorption while storing minerals and supporting load.

Control architecture. Mechanical strain, osteocytes, osteoblast/osteoclast coupling, hormones, calcium/phosphate, nutrition and age.

Failure modes. Low density/quality, fracture, excess remodeling, osteonecrosis or deformity.

Observable consequences. DXA, imaging, turnover markers, geometry, fracture history and functional loading.

Cross-system coupling. Endocrine, renal, hematologic, reproductive, aging and mobility systems.

Evidence anchors: [4,13,14]

3.0.2 Skeletal muscle excitation, force and adaptation

Core principle. Motor units convert neural activation and metabolic energy into force; fibers adapt to load, disuse, disease and aging.

Control architecture. Motor neuron recruitment, calcium, sarcomeres, mitochondria, protein turnover, hormones and training.

Failure modes. Weakness, myopathy, fatigue, atrophy, contracture, rhabdomyolysis or exercise intolerance.

Observable consequences. Strength, power, endurance, EMG, imaging, enzymes, biopsy and task performance.

Cross-system coupling. Nervous, endocrine, cardiovascular, respiratory and nutrition systems.

Evidence anchors: [3,14,15]

3.0.3 Cartilage, synovium and joint mechanics

Core principle. Low-friction cartilage, synovial fluid, capsule, ligaments and aligned bone distribute load through range of motion.

Control architecture. Matrix turnover, lubrication, mechanotransduction, neuromuscular control and inflammation.

Failure modes. Pain, instability, stiffness, degeneration, inflammatory synovitis or structural failure.

Observable consequences. Exam, motion, load response, imaging, fluid/tissue analysis and function.

Cross-system coupling. Immune, pain, obesity, occupation and rehabilitation.

Evidence anchors: [3,5,13]

3.0.4 Tendon, ligament and enthesis biology

Core principle. Dense collagen structures transmit muscle force and stabilize joints with slow adaptation and region-specific mechanics.

Control architecture. Strain magnitude/rate, matrix synthesis, vascularity, hormones, age and recovery.

Failure modes. Tendinopathy, tear, laxity, enthesitis, rupture or failed healing.

Observable consequences. Load-provoked symptoms, strength, ultrasound/MRI, motion and return-to-task.

Cross-system coupling. Muscle, bone, immune, occupation and sport.

Evidence anchors: [3,6]

3.0.5 Connective tissue and fascia

Core principle. Collagen, elastin, proteoglycans and fibroblasts organize force transmission, organ support and tissue repair.

Control architecture. Genetic matrix proteins, mechanical loading, inflammation, hormones and remodeling.

Failure modes. Hypermobility, fibrosis, fragility, contracture, scar or systemic connective-tissue disease.

Observable consequences. Phenotype, tissue mechanics, imaging, genetics/biopsy in selected cases and function.

Cross-system coupling. Skin, vascular, pulmonary, ocular and immune systems.

Evidence anchors: [6,13]

3.0.6 Spine and intervertebral system

Core principle. Vertebrae, discs, facets, ligaments, muscles and neural structures balance mobility, load and protection.

Control architecture. Posture, load distribution, disc nutrition, neuromuscular control, development and aging.

Failure modes. Axial pain, radiculopathy, myelopathy, stenosis, deformity, fracture or instability.

Observable consequences. Neurologic red flags, motion/function, imaging when indicated and psychosocial/work context.

Cross-system coupling. Nervous, occupational, pain, bone and rehabilitation systems.

Evidence anchors: [3,6,16]

3.0.7 Pain, nociception and protective behavior

Core principle. Nociceptors and central systems signal threat, but pain is an experience shaped by tissue input, sensitization, prediction, emotion and context.

Control architecture. Peripheral transduction, spinal/brain processing, descending modulation, learning, sleep, stress and social meaning.

Failure modes. Acute pain, persistent nociplastic pain, fear avoidance, hyperalgesia or loss of protective sensation.

Observable consequences. History, exam, function, patient report, sensory testing and mechanism-informed challenge.

Cross-system coupling. Brain/behavior, psychosocial, sleep, immune and occupational systems.

Evidence anchors: [10,11]

3.0.8 Movement control, balance and motor learning

Core principle. Sensation, prediction, motor planning, reflexes and musculoskeletal capacity produce efficient task-specific movement.

Control architecture. Central/peripheral nervous systems, vestibular/vision, proprioception, practice, fatigue and environment.

Failure modes. Falls, inefficient compensation, instability, learned nonuse or reinjury.

Observable consequences. Gait, kinematics/kinetics, balance, task performance, fatigue and participation.

Cross-system coupling. Nervous, sensory, cardiovascular, environment and assistive technology.

Evidence anchors: [7,9,15]

3.0.9 Repair, regeneration and fibrosis

Core principle. Injury triggers inflammation, proliferation, matrix deposition and remodeling with tissue-specific regenerative limits.

Control architecture. Cell source, vascular supply, mechanical environment, immune signals, nutrition and rehabilitation dose.

Failure modes. Nonunion, scar, adhesions, chronic inflammation, reinjury or pathologic ossification.

Observable consequences. Imaging, biomarkers, tissue mechanics, symptoms, load tolerance and return of function.

Cross-system coupling. Immune, vascular, metabolic and intervention systems.

Evidence anchors: [7,13]

3.0.10 Reserve across the life course

Core principle. Peak bone/muscle capacity, skill, activity and recovery shape resilience to injury, illness and aging.

Control architecture. Development, puberty, sex hormones, nutrition, physical activity, work, disease and senescence.

Failure modes. Frailty, sarcopenia, falls, fracture, disability and loss of independence.

Observable consequences. Body composition, strength, gait, activity, frailty and participation trajectories.

Cross-system coupling. Life course, endocrine, nutrition and social context.

Evidence anchors: [3,34,35]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Joint torque	$\tau = F r_{\perp}$	Force times perpendicular moment arm; internal tissue forces can greatly exceed external load.	Musculoskeletal systems have multiple muscles, changing geometry and dynamic terms.

Continued on next page

Relationship	Expression	Interpretation	Limits
Mechanical work and power	$W = \int \vec{F} \cdot d\vec{x}, \quad P = dW/dt$	Quantifies energy transfer and rate during movement.	Biological efficiency, elastic storage and co-contraction require broader models.
Stress and strain	$\sigma = F/A, \quad \varepsilon = \Delta L/L_0$	Normalizes force and deformation to geometry.	Tissues are anisotropic, viscoelastic, heterogeneous and history-dependent.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within musculoskeletal and connective-tissue health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within musculoskeletal and connective-tissue health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[21]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[35]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[3]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: ^[30]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[32]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the musculoskeletal disorder, tissue failure mode, pain mechanism, movement phenotype, or participation limitation, but cross-system processes and functional effects are retained.

5.0.1 Osteoarthritis

Definition and phenotype. Whole-joint disorder with cartilage loss, bone change, synovitis, pain and functional limitation.

Pathophysiology. Mechanical loading, aging, injury, metabolism and inflammation interact; structural severity and pain may diverge.

Determinants and modifiers. Age, prior injury, obesity, anatomy, occupation, activity, sex and social context.

Assessment. Clinical pattern and function; weight-bearing radiography or other imaging when diagnosis/management is uncertain.

Intervention logic. Education, exercise, weight management where relevant, topical/oral analgesia by risk, injections selectively and joint replacement for advanced disability.

Known limits and open questions. No disease-modifying drug reliably reverses established disease; imaging can drive overtreatment.

Suite links. 10 rehabilitation; 05 pain/behavior; 14 therapeutics.

Evidence anchors: [3,5,10]

5.0.2 Inflammatory arthritis

Definition and phenotype. Immune-mediated synovitis causing pain, stiffness, systemic effects and progressive joint damage.

Pathophysiology. Autoimmune/autoinflammatory pathways sustain synovial proliferation, cytokines and bone/cartilage erosion.

Determinants and modifiers. Genetics, smoking, infection, sex/hormones, obesity and diagnostic/treatment delay.

Assessment. Patterned exam, inflammatory markers/antibodies, ultrasound/imaging, organ comorbidity and function.

Intervention logic. Early treat-to-target disease-modifying therapy plus exercise, risk management and rehabilitation.

Known limits and open questions. Seronegative disease and mixed pain mechanisms complicate activity assessment.

Suite links. 03H immune; 14 pharmacology; 10 rehab.

Evidence anchors: ^[3,6]

5.0.3 Osteoporosis and fragility fracture

Definition and phenotype. Reduced bone strength leading to fractures from low-energy trauma.

Pathophysiology. Loss of density, microarchitecture and material quality combines with fall mechanics.

Determinants and modifiers. Age, menopause, low weight, inactivity, steroids, endocrine/renal disease, smoking/alcohol and prior fracture.

Assessment. Fracture history, DXA interpreted with clinical risk, secondary-cause evaluation, falls and vertebral imaging when indicated.

Intervention logic. Exercise/fall prevention, adequate nutrition, antiresorptive or anabolic treatment by risk, and fracture liaison systems.

Known limits and open questions. Risk tools and DXA incompletely capture bone quality and treatment sequencing/duration remain individualized.

Suite links. 03D endocrine; 03F renal; 07 aging; 10 falls.

Evidence anchors: ^[3,4,17]

5.0.4 Low back and neck pain

Definition and phenotype. Pain and disability arising from heterogeneous spinal, muscular, neural and psychosocial mechanisms, usually without one dangerous lesion.

Pathophysiology. Tissue irritation, muscle guarding, sensitization, nerve-root compression, expectations, work and sleep interact.

Determinants and modifiers. Workload, inactivity/deconditioning, prior episodes, mood, stress, smoking, obesity and social/occupational conditions.

Assessment. Red-flag and neurologic assessment, function and psychosocial factors; imaging only when it changes management.

Intervention logic. Stay active, education, exercise/rehabilitation, selected medications/procedures and targeted surgery for defined neurologic/structural disease.

Known limits and open questions. Incidental imaging is common and exact pain generators are often uncertain.

Suite links. 05 cognition/behavior; 06 psychosocial; 09 occupational; 10 rehab.

Evidence anchors: [3,6,10]

5.0.5 Tendinopathy and overuse injury

Definition and phenotype. Load-related tendon pain and impaired capacity with variable structural change.

Pathophysiology. Repeated or poorly tolerated loading alters matrix, cells, pain and motor behavior; inflammation is context-dependent.

Determinants and modifiers. Training/workload spikes, technique, age, metabolic disease, medications and recovery.

Assessment. Load history, localized exam, strength/function; ultrasound/MRI for uncertainty, tear or planning.

Intervention logic. Progressive loading and task modification; selected adjuncts; surgery for specific refractory structural disease.

Known limits and open questions. Imaging abnormalities occur without symptoms and biologic injections have inconsistent evidence.

Suite links. 09 occupation/sport; 10 rehabilitation; 14 interventions.

Evidence anchors: [3,5]

5.0.6 Acute fracture and dislocation

Definition and phenotype. Traumatic disruption of bone or joint alignment with potential neurovascular and soft-tissue injury.

Pathophysiology. Energy exceeds tissue tolerance; healing depends on stability, blood supply, biology and rehabilitation.

Determinants and modifiers. Falls, collision, sport/work, bone fragility, environment and violence.

Assessment. Immediate neurovascular/skin assessment, radiography/CT as needed and mechanism-specific evaluation.

Intervention logic. Reduction, immobilization or fixation, pain control, thrombosis/infection prevention and staged rehabilitation.

Known limits and open questions. Optimal fixation/rehabilitation differs by fracture, patient goals and system resources.

Suite links. 09 injury prevention; 10 rehab; 12 trauma systems.

Evidence anchors: [3,32]

5.0.7 Muscle strain, tear and rhabdomyolysis

Definition and phenotype. Muscle injury ranges from focal fiber disruption to widespread breakdown releasing intracellular contents.

Pathophysiology. Excess eccentric load, ischemia, heat, toxins, seizures or metabolic defects damage sarcolemma and energy balance.

Determinants and modifiers. Training, heat, drugs, infection, inherited metabolism and delayed recognition.

Assessment. History/exam, CK/renal/electrolytes when severe, imaging for grade/complication and function.

Intervention logic. Remove cause, protect then reload focal injuries; urgent fluid/electrolyte/renal management for rhabdomyolysis according to severity.

Known limits and open questions. CK poorly predicts all outcomes and return-to-play thresholds are not universal.

Suite links. 03F renal; 03O homeostasis; 09 heat/occupation.

Evidence anchors: [3,14,18]

5.0.8 Sarcopenia and frailty-related weakness

Definition and phenotype. Loss of muscle strength and performance, often with reduced mass, contributing to vulnerability and disability.

Pathophysiology. Aging, inactivity, inflammation, denervation, endocrine change, malnutrition and disease reduce anabolic response and reserve.

Determinants and modifiers. Age, hospitalization, social isolation, low activity, poor nutrition, multimorbidity and medications.

Assessment. Strength and performance first, body composition where useful, nutrition/disease review and participation.

Intervention logic. Progressive resistance/power exercise, protein/energy adequacy, disease/medication optimization and fall prevention.

Known limits and open questions. Definitions and cutoffs vary; mass alone misses function.

Suite links. 07 aging; 01 whole-person; 10 rehabilitation.

Evidence anchors: [3,7,34]

5.0.9 Inflammatory and inherited myopathy

Definition and phenotype. Muscle weakness from immune, genetic, metabolic, endocrine, toxic or infectious mechanisms.

Pathophysiology. Fiber necrosis, inflammation, protein/energy defects, mitochondrial failure or membrane instability.

Determinants and modifiers. Genotype, autoimmunity, drugs, endocrine disease, infection and exertion.

Assessment. Patterned strength, CK and labs, EMG, MRI, antibodies/genetics and biopsy selectively.

Intervention logic. Cause-specific immunotherapy, metabolic or supportive therapy plus rehabilitation and cardiopulmonary surveillance.

Known limits and open questions. Biomarkers and biopsy can be nondiagnostic; many rare variants remain uncertain.

Suite links. 03C nervous; 03D endocrine; 03H immune; 03A/03B surveillance.

Evidence anchors: [6,16,21]

5.0.10 Connective-tissue disorders and hypermobility phenotypes

Definition and phenotype. Inherited or acquired matrix disorders causing joint laxity, tissue fragility or multisystem features.

Pathophysiology. Collagen/elastin/ECM variants or immune fibrosis alter mechanical properties across tissues.

Determinants and modifiers. Genotype, sex/hormones, injury, conditioning, pain sensitization and diagnostic labeling.

Assessment. Standardized phenotype, family history, targeted genetics and organ screening based on syndrome; avoid overinterpreting nonspecific flexibility.

Intervention logic. Protection from high-risk stress, individualized strengthening/proprioception, symptom management and syndrome-specific surveillance.

Known limits and open questions. Broad hypermobility labels mix distinct biology and evidence for many interventions is limited.

Suite links. 03A vascular; 03J skin; 03K eye; 05 pain; 15 diagnostic equity.

Evidence anchors: [10,13,21]

5.0.11 Septic arthritis and osteomyelitis

Definition and phenotype. Infection of joint or bone causing rapid tissue damage and systemic risk.

Pathophysiology. Hematogenous spread, direct inoculation or contiguous infection drives inflammation, necrosis and biofilm.

Determinants and modifiers. Age, diabetes, vascular disease, surgery/devices, injection, wounds and immune suppression.

Assessment. Urgent aspiration/culture or imaging/tissue sampling plus systemic assessment; do not rely on serum markers alone.

Intervention logic. Rapid source drainage/debridement and targeted antimicrobials; stabilize and rehabilitate after control.

Known limits and open questions. Culture-negative disease and biofilm complicate diagnosis and duration.

Suite links. 08 infection; 03H immune; 14 antimicrobials.

Evidence anchors: [6,20,32]

5.0.12 Amputation and limb difference

Definition and phenotype. Congenital or acquired loss/difference of limb structure with effects determined by level, tissue health, pain, device fit and environment.

Pathophysiology. Trauma, vascular disease, infection, cancer or development alters mechanics, sensation and participation.

Determinants and modifiers. Prevention access, diabetes/vascular disease, work/traffic injury, conflict, rehabilitation and device availability.

Assessment. Residual-limb/skin, pain, strength, balance, energy cost, prosthetic goals and environment.

Intervention logic. Cause prevention, surgery/wound care, pain treatment, rehabilitation, prosthetics/orthotics and participation support.

Known limits and open questions. Technology performance in laboratories may not translate to daily use or affordability.

Suite links. 10 rehab/assistive technology; 09 injury; 15 rights/equity.

Evidence anchors: [7,8,9]

5.0.13 Chronic widespread and nociplastic pain

Definition and phenotype. Persistent pain with altered nociceptive processing and multidomain symptoms not fully explained by local tissue injury.

Pathophysiology. Central/peripheral sensitization, sleep, stress, autonomic, immune and learning processes interact with context.

Determinants and modifiers. Prior pain/injury, trauma/stress, sleep, mood, inequity, activity cycling and healthcare experiences.

Assessment. Positive multidimensional assessment, exclusion of urgent/treatable disease, function, sleep, mood and goals; avoid endless low-yield imaging.

Intervention logic. Education, graded individualized activity, psychological/behavioral and sleep interventions, selected medications and coordinated care.

Known limits and open questions. No single biomarker establishes nociplastic pain; heterogeneity and stigma remain substantial.

Suite links. 05 brain/behavior; 06 psychosocial; 10 rehab; 01 whole-person.

Evidence anchors: [10,11,12]

5.0.14 Bone and soft-tissue tumors

Definition and phenotype. Benign or malignant neoplasms affecting bone, muscle, connective tissue or nerves with local and systemic consequences.

Pathophysiology. Driver alterations, growth, invasion and metastasis disrupt mechanical integrity and function.

Determinants and modifiers. Age, inherited syndromes, radiation and mostly unknown causes.

Assessment. Imaging with tissue-pathology planning by specialist team; avoid unplanned excision of suspicious masses.

Intervention logic. Surgery, radiation and systemic/targeted therapy plus reconstruction and rehabilitation.

Known limits and open questions. Rare subtypes limit trial evidence and molecular findings do not always imply benefit.

Suite links. 14 oncology; 10 rehabilitation; 15 access.

Evidence anchors: ^[19,21]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current musculoskeletal and connective-tissue health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

Continued on next page

Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current musculoskeletal and connective-tissue health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[9,11]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[13]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.
Evidence anchors: ^[9]

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
History, physical examination and functional task testing	Noninvasive	Symptoms, red flags, palpation, range, strength, neurologic/vascular assessment and standardized tasks.	Mechanism hypotheses, urgency, baseline capability and treatment targets.	Low cost and directly linked to function.	Examiner variation, pain/effort, context and limited structural specificity.
Plain radiography and fluoroscopy	Noninvasive ionizing imaging	X-ray attenuation, static or dynamic projection.	Fracture, alignment, joint space, mineralization, hardware and motion.	Accessible and high value for many structural questions.	Radiation, projection overlap and poor soft-tissue sensitivity.
CT and quantitative CT	Noninvasive ionizing imaging	Cross-sectional x-ray attenuation with reconstruction.	Complex bone anatomy, fracture, mineral density proxy, surgical planning and some soft tissue.	High spatial resolution and fast acquisition.	Radiation, artifacts and incidental findings.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
MRI	Noninvasive	Nuclear magnetic resonance signal sensitive to water, fat, flow and tissue microenvironment.	Marrow, cartilage, disc, muscle, tendon, ligament, tumor, inflammation and neural structures.	Excellent soft-tissue contrast without ionizing radiation.	Cost, time, motion, implants, nonspecific signal and incidental findings.
Musculoskeletal ultrasound	Noninvasive	Reflected high-frequency sound, Doppler and dynamic scanning.	Tendon/muscle, synovium, effusion, superficial nerve, motion and guided procedures.	Portable, dynamic and relatively low cost.	Operator dependence, limited deep field and anisotropy artifacts.
DXA and bone-density measurement	Low-dose ionizing imaging	Dual-energy attenuation estimates areal bone mineral density and body composition.	T-score/Z-score, regional density and selected composition.	Standardized fracture-risk input and monitoring.	Two-dimensional, size/degeneration artifacts and small change relative to precision.
Force platforms and dynamometry	Noninvasive	Load cells, torque transducers and controlled movement.	Strength, power, balance, asymmetry and rate of force development.	Quantitative functional capacity.	Effort, learning, positioning and device-specific norms.
Motion capture and inertial sensing	Noninvasive	Optical markers/depth video or accelerometer/gyroscope/magnetometer signals.	Kinematics, gait, activity, joint angles and task strategy.	Laboratory precision or real-world longitudinal coverage.	Soft-tissue artifact, drift, occlusion, algorithms and environment.
Surface and needle electromyography	Noninvasive or minimally invasive	Electrical potentials from muscle fibers and motor units.	Activation timing, recruitment, conduction and neuromuscular patterns.	Direct electrical window on muscle activation.	Cross-talk, electrode geometry, adiposity, effort and nonspecific abnormalities.
Gait and plantar-pressure systems	Noninvasive	Force, pressure, timing and motion sensors in platforms, insoles or wearables.	Step timing, loading, pressure risk, symmetry and mobility.	Links mechanics to walking and footwear/prostheses.	Calibration drift, footwear/context and algorithm dependence.
Quantitative sensory testing	Noninvasive	Controlled thermal, mechanical, vibration or electrical stimuli with responses.	Detection/pain thresholds and sensory profiles.	Mechanism clues for neuropathic/nociplastic states.	Attention, expectation, protocol and poor diagnostic specificity alone.

Continued on next page

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Arthrocentesis and synovial analysis	Minimally invasive	Fluid aspiration with cell count, crystals, culture and other assays.	Infection, crystals, inflammation and effusion characterization.	Can rapidly change urgent treatment.	Sampling error, contamination, false-negative cultures and procedural risk.
Bone, muscle, synovial or tumor biopsy	Invasive	Histology, microbiology, biochemical and molecular tissue analysis.	Inflammation, infection, malignancy, myopathy and metabolic bone pathology.	Direct tissue evidence.	Sampling, harm and treatment/timing effects.
Wearable activity and load monitoring	Noninvasive	Inertial, pressure, heart-rate and context sensors.	Activity dose, sedentary time, sleep/activity cycles, loading and recovery proxies.	Longitudinal real-world behavior.	Nonwear, algorithm/device drift, disability bias and quantity-quality confusion.
Elastography and tissue mechanical imaging	Noninvasive	Ultrasound or MR response to induced/internal shear waves.	Relative stiffness of muscle, tendon, fascia and lesions.	Potential quantitative tissue mechanics.	Anisotropy, preload, depth, vendor methods and uncertain clinical thresholds.
Smart prosthetic, orthotic and rehabilitation robotics sensing	Noninvasive human-device interface	Force, position, EMG, pressure, inertial and sometimes neural signals.	Device state, intent, fit, loading and training feedback.	Can personalize assistance and quantify use.	Skin injury, control instability, cybersecurity, abandonment and inequitable access.

7.0.1 History, physical examination and functional task testing

Measured quantity and transducer. Symptoms, red flags, palpation, range, strength, neurologic/vascular assessment and standardized tasks.

Typical outputs. Mechanism hypotheses, urgency, baseline capability and treatment targets.

Strengths. Low cost and directly linked to function.

Limitations and artifacts. Examiner variation, pain/effort, context and limited structural specificity.

Reference and validation strategy. Standard protocols, reliability and outcome responsiveness.

Clinical/research maturity. Established core.

Evidence anchors: [3,9]

7.0.2 Plain radiography and fluoroscopy

Measured quantity and transducer. X-ray attenuation, static or dynamic projection.

Typical outputs. Fracture, alignment, joint space, mineralization, hardware and motion.

Strengths. Accessible and high value for many structural questions.

Limitations and artifacts. Radiation, projection overlap and poor soft-tissue sensitivity.

Reference and validation strategy. Dose/geometry controls and indication-specific reference interpretation.

Clinical/research maturity. Established.

Evidence anchors: [6,32]

7.0.3 CT and quantitative CT

Measured quantity and transducer. Cross-sectional x-ray attenuation with reconstruction.

Typical outputs. Complex bone anatomy, fracture, mineral density proxy, surgical planning and some soft tissue.

Strengths. High spatial resolution and fast acquisition.

Limitations and artifacts. Radiation, artifacts and incidental findings.

Reference and validation strategy. Protocol/dose control, calibration phantoms and task-specific validation.

Clinical/research maturity. Established.

Evidence anchors: [6,22]

7.0.4 MRI

Measured quantity and transducer. Nuclear magnetic resonance signal sensitive to water, fat, flow and tissue microenvironment.

Typical outputs. Marrow, cartilage, disc, muscle, tendon, ligament, tumor, inflammation and neural structures.

Strengths. Excellent soft-tissue contrast without ionizing radiation.

Limitations and artifacts. Cost, time, motion, implants, nonspecific signal and incidental findings.

Reference and validation strategy. Standard sequences, safety screening and clinical-pathology/operative correlation.

Clinical/research maturity. Established.

Evidence anchors: [6,32]

7.0.5 Musculoskeletal ultrasound

Measured quantity and transducer. Reflected high-frequency sound, Doppler and dynamic scanning.

Typical outputs. Tendon/muscle, synovium, effusion, superficial nerve, motion and guided procedures.

Strengths. Portable, dynamic and relatively low cost.

Limitations and artifacts. Operator dependence, limited deep field and anisotropy artifacts.

Reference and validation strategy. Training, phantom, inter-rater and MRI/operative comparison as appropriate.

Clinical/research maturity. Established.

Evidence anchors: [6,22]

7.0.6 DXA and bone-density measurement

Measured quantity and transducer. Dual-energy attenuation estimates areal bone mineral density and body composition.

Typical outputs. T-score/Z-score, regional density and selected composition.

Strengths. Standardized fracture-risk input and monitoring.

Limitations and artifacts. Two-dimensional, size/degeneration artifacts and small change relative to precision.

Reference and validation strategy. Phantom QA, least significant change, positioning and fracture-outcome calibration.

Clinical/research maturity. Established.

Evidence anchors: [4,17]

7.0.7 Force platforms and dynamometry

Measured quantity and transducer. Load cells, torque transducers and controlled movement.

Typical outputs. Strength, power, balance, asymmetry and rate of force development.

Strengths. Quantitative functional capacity.

Limitations and artifacts. Effort, learning, positioning and device-specific norms.

Reference and validation strategy. Calibration weights/arms, repeatability and task/outcome linkage.

Clinical/research maturity. Established to research.

Evidence anchors: [7,9]

7.0.8 Motion capture and inertial sensing

Measured quantity and transducer. Optical markers/depth video or accelerometer/gyroscope/magnetometer signals.

Typical outputs. Kinematics, gait, activity, joint angles and task strategy.

Strengths. Laboratory precision or real-world longitudinal coverage.

Limitations and artifacts. Soft-tissue artifact, drift, occlusion, algorithms and environment.

Reference and validation strategy. Timing/calibration, reference motion system, cross-task and real-world validation.

Clinical/research maturity. Established research; growing clinical use.

Evidence anchors: [22,29]

7.0.9 Surface and needle electromyography

Measured quantity and transducer. Electrical potentials from muscle fibers and motor units.

Typical outputs. Activation timing, recruitment, conduction and neuromuscular patterns.

Strengths. Direct electrical window on muscle activation.

Limitations and artifacts. Cross-talk, electrode geometry, adiposity, effort and nonspecific abnormalities.

Reference and validation strategy. Signal standards, reference tasks and clinical electrodiagnostic correlation.

Clinical/research maturity. Established specialist/research.

Evidence anchors: [16,27]

7.0.10 Gait and plantar-pressure systems

Measured quantity and transducer. Force, pressure, timing and motion sensors in platforms, insoles or wearables.

Typical outputs. Step timing, loading, pressure risk, symmetry and mobility.

Strengths. Links mechanics to walking and footwear/prostheses.

Limitations and artifacts. Calibration drift, footwear/context and algorithm dependence.

Reference and validation strategy. Traceable loads, reference walkway and free-living outcome studies.

Clinical/research maturity. Established to emerging.

Evidence anchors: [7,8]

7.0.11 Quantitative sensory testing

Measured quantity and transducer. Controlled thermal, mechanical, vibration or electrical stimuli with responses.

Typical outputs. Detection/pain thresholds and sensory profiles.

Strengths. Mechanism clues for neuropathic/nociplastic states.

Limitations and artifacts. Attention, expectation, protocol and poor diagnostic specificity alone.

Reference and validation strategy. Standardized methods, repeatability and outcome-linked interpretation.

Clinical/research maturity. Research/specialist.

Evidence anchors: [10,16]

7.0.12 Arthrocentesis and synovial analysis

Measured quantity and transducer. Fluid aspiration with cell count, crystals, culture and other assays.

Typical outputs. Infection, crystals, inflammation and effusion characterization.

Strengths. Can rapidly change urgent treatment.

Limitations and artifacts. Sampling error, contamination, false-negative cultures and procedural risk.

Reference and validation strategy. Prompt handling, microscopy/culture QA and clinical follow-up.

Clinical/research maturity. Established.

Evidence anchors: [6,20]

7.0.13 Bone, muscle, synovial or tumor biopsy

Measured quantity and transducer. Histology, microbiology, biochemical and molecular tissue analysis.

Typical outputs. Inflammation, infection, malignancy, myopathy and metabolic bone pathology.

Strengths. Direct tissue evidence.

Limitations and artifacts. Sampling, harm and treatment/timing effects.

Reference and validation strategy. Expert integrated pathology and planned trajectory to avoid compromising surgery.

Clinical/research maturity. Established in selected indications.

Evidence anchors: [13,19]

7.0.14 Wearable activity and load monitoring

Measured quantity and transducer. Inertial, pressure, heart-rate and context sensors.

Typical outputs. Activity dose, sedentary time, sleep/activity cycles, loading and recovery proxies.

Strengths. Longitudinal real-world behavior.

Limitations and artifacts. Nonwear, algorithm/device drift, disability bias and quantity-quality confusion.

Reference and validation strategy. Raw/quality data, task-specific reference and utility trials.

Clinical/research maturity. Emerging.

Evidence anchors: [22,29]

7.0.15 Elastography and tissue mechanical imaging

Measured quantity and transducer. Ultrasound or MR response to induced/internal shear waves.

Typical outputs. Relative stiffness of muscle, tendon, fascia and lesions.

Strengths. Potential quantitative tissue mechanics.

Limitations and artifacts. Anisotropy, preload, depth, vendor methods and uncertain clinical thresholds.

Reference and validation strategy. Phantoms, protocol control, repeatability and outcome-linked validation.

Clinical/research maturity. Emerging/research.

Evidence anchors: [22,25]

7.0.16 Smart prosthetic, orthotic and rehabilitation robotics sensing

Measured quantity and transducer. Force, position, EMG, pressure, inertial and sometimes neural signals.

Typical outputs. Device state, intent, fit, loading and training feedback.

Strengths. Can personalize assistance and quantify use.

Limitations and artifacts. Skin injury, control instability, cybersecurity, abandonment and inequitable access.

Reference and validation strategy. Bench safety, human factors, long-term free-living use and patient-important outcomes.

Clinical/research maturity. Established components; advanced control emerging.

Evidence anchors: [8,24,27,28]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise musculoskeletal and connective-tissue health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Continued on next page

Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise musculoskeletal and connective-tissue health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [22,26]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [26,27]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[25]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[25]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[29]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[29]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[24,27,28]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Physical activity, strength and motor training	Prevention/rehabilitation	Build functional capacity, motor skill, metabolic health and confidence through progressive dose.	Strong for many pain, arthritis, frailty, osteoporosis and recovery outcomes when individualized.	Overload, falls, flare, poor accessibility and adherence burden.	Symptoms, load tolerance, strength, function, participation and adverse events.
Education, self-management and work/activity modification	Whole-person/system	Reduce fear, improve pacing/load decisions, ergonomics and shared goals.	Useful across chronic musculoskeletal conditions as part of active care.	Messages can blame patients or promote avoidance when poorly designed.	Knowledge, behavior, work/role participation, flare pattern and treatment burden.
Analgesic and anti-inflammatory pharmacotherapy	Symptom/disease treatment	Modulate inflammatory mediators, neural signaling or central pain processing.	Benefits are usually modest to moderate and condition-specific; disease-modifying agents differ from symptom drugs.	GI, renal, cardiovascular, sedation, dependence, interactions and falls.	Pain/function, target attainment, organ safety, use pattern and deprescribing opportunities.
Disease-modifying antirheumatic therapy	Disease modification	Suppress specific or broad immune pathways to prevent inflammatory damage.	Strong in defined inflammatory arthritides with early treat-to-target strategy.	Infection, organ toxicity, malignancy signals, pregnancy constraints and cost.	Disease activity, imaging/structure when needed, labs, infection and function.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Osteoporosis pharma-cotherapy	Fracture prevention	Reduce resorption or stimulate formation according to fracture risk and prior therapy.	Reduces selected fracture outcomes in high-risk groups.	Rare skeletal effects, renal limits, rebound after some agents and sequencing complexity.	Adherence, calcium/vitamin status, fractures, DXA at meaningful intervals and adverse effects.
Injections and image-guided procedures	Symptom/diagnosis	Deliver anesthetic, corticosteroid or other agents; aspirate or target specific structures.	Useful for selected short-term indications and diagnostic questions.	Infection, tissue injury, systemic effects, repeated-use harm and variable biologic-product evidence.	Response, function, adverse events and downstream surgery timing.
Surgery and fixation/reconstruction	Structural treatment	Restore alignment, stability, decompression, replace joints or remove diseased tissue.	Large benefit for selected fracture, advanced joint disease, compression, tumor or deformity.	Infection, thrombosis, neurovascular injury, revision, persistent pain and rehabilitation burden.	Complications, alignment/union, pain, function, participation and device longevity.
Orthoses, prostheses and assistive products	Assistive/rehabilitation	Redistribute load, stabilize, substitute structure or augment task performance.	Can improve mobility and participation when fitted to person and environment.	Skin injury, discomfort, abandonment, maintenance and cost.	Fit, wear time, skin, energy cost, task outcomes and user goals.
Fall and injury prevention	Prevention/systems	Address strength/balance, vision, medicines, environment, work/sport hazards and bone fragility.	Multifactorial approaches reduce risk in selected populations.	Intervention burden, poor targeting and inequitable environmental control.	Falls/injury, near falls, strength/balance, hazards and adherence.
Psychological and behavioral pain interventions	Whole-person	Alter coping, fear avoidance, attention, sleep, mood and activity patterns without denying biological pain.	Evidence supports selected cognitive-behavioral, acceptance and multidisciplinary approaches.	Stigma, access and misuse as a substitute for needed medical evaluation.	Function, distress, pain interference, sleep, activity and goals.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Regenerative, orthobiologic and tissue-engineering interventions	Frontier	Attempt to alter repair using cells, growth factors, scaffolds or biologic preparations.	Established only for narrow products/indications; many marketed uses lack high-quality comparative evidence.	Infection, ectopic tissue, immune effects, manufacturing variability, cost and false claims.	Product characterization, imaging/tissue, function, adverse events and long-term trials.
Multidisciplinary rehabilitation and vocational reintegration	Coordinate	Coordinate physical, occupational, psychological, social and workplace interventions around participation goals.	Important for complex injury, chronic pain, amputation and multimorbidity.	Fragmentation, insurer barriers and poor outcome alignment.	Goal attainment, independence, return to roles/work, quality of life and sustained use.

9.0.1 Physical activity, strength and motor training

Target and mechanism. Build tissue capacity, motor skill, metabolic health and confidence through progressive dose.

Evidence and expected outcome. Strong for many pain, arthritis, frailty, osteoporosis and recovery outcomes when individualized.

Risks, burden, and equity. Overload, falls, flare, poor accessibility and adherence burden.

Response monitoring. Symptoms, load tolerance, strength, function, participation and adverse events.

Evidence anchors: [3,7]

9.0.2 Education, self-management and work/activity modification

Target and mechanism. Reduce fear, improve pacing/load decisions, ergonomics and shared goals.

Evidence and expected outcome. Useful across chronic musculoskeletal conditions as part of active care.

Risks, burden, and equity. Messages can blame patients or promote avoidance when poorly designed.

Response monitoring. Knowledge, behavior, work/role participation, flare pattern and treatment burden.

Evidence anchors: [9,30,33]

9.0.3 Analgesic and anti-inflammatory pharmacotherapy

Target and mechanism. Modulate inflammatory mediators, neural signaling or central pain processing.

Evidence and expected outcome. Benefits are usually modest to moderate and condition-specific; disease-modifying agents differ from symptom drugs.

Risks, burden, and equity. GI, renal, cardiovascular, sedation, dependence, interactions and falls.

Response monitoring. Pain/function, target attainment, organ safety, use pattern and deprescribing opportunities.

Evidence anchors: [6,10,23]

9.0.4 Disease-modifying antirheumatic therapy

Target and mechanism. Suppress specific or broad immune pathways to prevent inflammatory damage.

Evidence and expected outcome. Strong in defined inflammatory arthritides with early treat-to-target strategy.

Risks, burden, and equity. Infection, organ toxicity, malignancy signals, pregnancy constraints and cost.

Response monitoring. Disease activity, imaging/structure when needed, labs, infection and function.

Evidence anchors: [6,21]

9.0.5 Osteoporosis pharmacotherapy

Target and mechanism. Reduce resorption or stimulate formation according to fracture risk and prior therapy.

Evidence and expected outcome. Reduces selected fracture outcomes in high-risk groups.

Risks, burden, and equity. Rare skeletal effects, renal limits, rebound after some agents and sequencing complexity.

Response monitoring. Adherence, calcium/vitamin status, fractures, DXA at meaningful intervals and adverse effects.

Evidence anchors: [4,17]

9.0.6 Injections and image-guided procedures

Target and mechanism. Deliver anesthetic, corticosteroid or other agents; aspirate or target specific structures.

Evidence and expected outcome. Useful for selected short-term indications and diagnostic questions.

Risks, burden, and equity. Infection, tissue injury, systemic effects, repeated-use harm and variable biologic-product evidence.

Response monitoring. Response, function, adverse events and downstream surgery timing.

Evidence anchors: [5,32]

9.0.7 Surgery and fixation/reconstruction

Target and mechanism. Restore alignment, stability, decompression, replace joints or remove diseased tissue.

Evidence and expected outcome. Large benefit for selected fracture, advanced joint disease, compression, tumor or deformity.

Risks, burden, and equity. Infection, thrombosis, neurovascular injury, revision, persistent pain and rehabilitation burden.

Response monitoring. Complications, alignment/union, pain, function, participation and device longevity.

Evidence anchors: [3,32]

9.0.8 Orthoses, prostheses and assistive products

Target and mechanism. Redistribute load, stabilize, substitute structure or augment task performance.

Evidence and expected outcome. Can improve mobility and participation when fitted to person and environment.

Risks, burden, and equity. Skin injury, discomfort, abandonment, maintenance and cost.

Response monitoring. Fit, wear time, skin, energy cost, task outcomes and user goals.

Evidence anchors: [7,8]

9.0.9 Fall and injury prevention

Target and mechanism. Address strength/balance, vision, medicines, environment, work/sport hazards and bone fragility.

Evidence and expected outcome. Multifactorial approaches reduce risk in selected populations.

Risks, burden, and equity. Intervention burden, poor targeting and inequitable environmental control.

Response monitoring. Falls/injury, near falls, strength/balance, hazards and adherence.

Evidence anchors: [3,30,31]

9.0.10 Psychological and behavioral pain interventions

Target and mechanism. Alter coping, fear avoidance, attention, sleep, mood and activity patterns without denying biological pain.

Evidence and expected outcome. Evidence supports selected cognitive-behavioral, acceptance and multidisciplinary approaches.

Risks, burden, and equity. Stigma, access and misuse as a substitute for needed medical evaluation.

Response monitoring. Function, distress, pain interference, sleep, activity and goals.

Evidence anchors: [10,11,12]

9.0.11 Regenerative, orthobiologic and tissue-engineering interventions

Target and mechanism. Attempt to alter repair using cells, growth factors, scaffolds or biologic preparations.

Evidence and expected outcome. Established only for narrow products/indications; many marketed uses lack high-quality comparative evidence.

Risks, burden, and equity. Infection, ectopic tissue, immune effects, manufacturing variability, cost and false claims.

Response monitoring. Product characterization, imaging/tissue, function, adverse events and long-term trials.

Evidence anchors: [21,23]

9.0.12 Multidisciplinary rehabilitation and vocational reintegration

Target and mechanism. Coordinate physical, occupational, psychological, social and workplace interventions around participation goals.

Evidence and expected outcome. Important for complex injury, chronic pain, amputation and multimorbidity.

Risks, burden, and equity. Fragmentation, insurer barriers and poor outcome alignment.

Response monitoring. Goal attainment, independence, return to roles/work, quality of life and sustained use.

Evidence anchors: [7,9,33]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can imaging biomarkers predict pain and recovery?	Imaging detects many structural and inflammatory features.	Individual causal contribution and treatment responsiveness for common pain.	Abnormalities are prevalent without symptoms and timing differs.	Prospective challenge/intervention studies with function and blinded imaging.
Which wearable load metrics prevent injury?	Wearables quantify selected motion and external load.	Transferable thresholds that improve outcomes without reducing beneficial activity.	Internal tissue load, adaptation and context are weakly inferred.	Pragmatic randomized feedback trials with injury, performance and behavior outcomes.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within musculoskeletal and connective-tissue health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Continued on next page

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can imaging biomarkers predict pain and recovery?

Established knowledge. Imaging detects many structural and inflammatory features.

Unresolved issue. Individual causal contribution and treatment responsiveness for common pain.

Why current evidence is insufficient. Abnormalities are prevalent without symptoms and timing differs.

Research that would change confidence. Prospective challenge/intervention studies with function and blinded imaging.

Evidence anchors: ^[6,10]

10.0.2 Which wearable load metrics prevent injury?

Established knowledge. Wearables quantify selected motion and external load.

Unresolved issue. Transferable thresholds that improve outcomes without reducing beneficial activity.

Why current evidence is insufficient. Internal tissue load, adaptation and context are weakly inferred.

Research that would change confidence. Pragmatic randomized feedback trials with injury, performance and behavior outcomes.

Evidence anchors: [22,29]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within musculoskeletal and connective-tissue health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: [21]

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [11,29]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[19]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[22]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[9,11]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Structure-pain-function cohort	Which combinations of tissue, sensory, motor and psychosocial states define actionable phenotypes?	Longitudinal multimodal cohort with standardized imaging, mechanics, sensory tests, PROMs, work/context and treatment response.	Externally validated treatment-interaction or prognosis improvement.	Clusters that are unstable or do not change decisions.
0. Open rehabilitation sensor stack	Can calibrated low-cost force, IMU, EMG and pressure sensing quantify recovery at home?	Bench metrology, synchronized reference lab comparison, diverse disability cohort and pragmatic utility trial.	Reliable quality-aware metrics that improve individualized rehabilitation.	Algorithmic scores drift across tasks, devices or assistive-product users.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for musculoskeletal and connective-tissue health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

Continued on next page

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Structure-pain-function cohort

Question. Which combinations of tissue, sensory, motor and psychosocial states define actionable phenotypes?

Design. Longitudinal multimodal cohort with standardized imaging, mechanics, sensory tests, PROMs, work/context and treatment response.

Success criterion. Externally validated treatment-interaction or prognosis improvement.

Failure or stop criterion. Clusters that are unstable or do not change decisions.

Evidence anchors: ^[10]

11.0.2 0. Open rehabilitation sensor stack

Question. Can calibrated low-cost force, IMU, EMG and pressure sensing quantify recovery at home?

Design. Bench metrology, synchronized reference lab comparison, diverse disability cohort and pragmatic utility trial.

Success criterion. Reliable quality-aware metrics that improve individualized rehabilitation.

Failure or stop criterion. Algorithmic scores drift across tasks, devices or assistive-product users.

Evidence anchors: ^[7,8,22]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for musculoskeletal and connective-tissue health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[26,29]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[33]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: ^[9]

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[33]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03C	Nervous System Health	Motor control, sensation, neuropathy, radiculopathy and neuromuscular disease shape movement.	weakness, pain or gait impairment may be neural or mixed.
03H	Immune and Lymphatic Health	Inflammatory arthritis, connective-tissue disease and immune therapy.	inflammation or autoimmunity is part of the phenotype.
05	Brain, Cognition, Emotion and Behavior	Pain processing, fear, learning, motivation, sleep and behavior influence disability and recovery.	symptom burden and tissue findings diverge.
09	Environmental and Occupational Health	Workload, ergonomics, injury, heat and built environment determine exposure and participation.	prevention or return-to-work is central.
10	Function, Disability and Rehabilitation	Activity, participation, rehabilitation, prosthetics and assistive technology are core outcomes.	moving from diagnosis to lived capability.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Tissue capacity	Load, motion and recovery that a tissue or system can tolerate under stated conditions.
Nociplastic pain	Pain arising from altered nociception despite no clear evidence that ongoing tissue damage or somatosensory lesion fully explains it.
Kinematics	Description of movement without regard to forces.
Kinetics	Forces and moments associated with movement.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

- [1] Technical Evidence and Systems Review. Atlas of Human Health: A Unified Taxonomy. 2026. Master taxonomy and conceptual architecture for this suite.
- [2] Technical Evidence and Engineering Review. Cardiovascular Health and Measurement Engineering. 2026. Volume 03A of this suite.
- [3] World Health Organization. Musculoskeletal health. 2025. <https://www.who.int/news-room/fact-sheets/detail/musculoskeletal-conditions>
- [4] International Osteoporosis Foundation. Clinical resources and fracture risk assessment. 2026. <https://www.osteoporosis.foundation/health-professionals>
- [5] American College of Rheumatology and Arthritis Foundation. Guideline for management of osteoarthritis. 2019. <https://rheumatology.org/osteoarthritis-guideline>
- [6] American College of Rheumatology. Clinical practice guidelines. 2026. <https://rheumatology.org/clinical-practice-guidelines>
- [7] World Health Organization. Rehabilitation 2030 initiative. 2026. <https://www.who.int/initiatives/rehabilitation-2030>
- [8] World Health Organization and UNICEF. Global report on assistive technology. 2022/2026. <https://www.who.int/publications/i/item/9789240049451>
- [9] World Health Organization. International Classification of Functioning, Disability and Health. 2001/2026. <https://www.who.int/standards/classifications/international-classification-of-functioning>
- [10] International Association for the Study of Pain. IASP terminology and ICD-11 pain classification. 2025/2026. <https://www.iasp-pain.org/resources/terminology/>
- [11] National Institutes of Health. Patient-Reported Outcomes Measurement Information System (PROMIS). 2026. <https://commonfund.nih.gov/promis>
- [12] World Health Organization. WHOQOL: Measuring Quality of Life. 2026. <https://www.who.int/tools/whoqol>
- [13] Kumar V, Abbas AK, Aster JC. Robbins and Cotran Pathologic Basis of Disease, 11th ed.. 2025.
- [14] Hall JE, Hall ME. Guyton and Hall Textbook of Medical Physiology, 15th ed.. 2025.
- [15] Movement Disorder Society. Clinical resources and rating scales. 2026. <https://www.movementdisorders.org/MDS/Resources/Clinical-Resources.htm>
- [16] American Academy of Neurology. Clinical practice guidelines. 2026. <https://www.aan.com/Guidelines/home>
- [17] Endocrine Society. Clinical practice guidelines. 2026. <https://www.endocrine.org/clinical-practice-guidelines>
- [18] Kidney Disease: Improving Global Outcomes. KDIGO Clinical Practice Guideline for Acute Kidney Injury. 2012/updates. <https://kdigo.org/guidelines/acute-kidney-injury/>

- [19] National Cancer Institute. Cancer biology, screening, diagnosis and treatment resources. 2026. <https://www.cancer.gov/>
- [20] World Health Organization. Global Action Plan on Antimicrobial Resistance 2026-2036. 2026. <https://www.who.int/news/item/25-05-2026-the-world-health-assembly-adopts-updated-global-action-plan-2026-2036>
- [21] World Health Organization. Precision medicine: a path towards targeted, personalized and equitable care (WHA79.8). 2026. <https://www.who.int/news/item/22-05-2026-world-health-assembly-endorses-precision-medicine>
- [22] U.S. Food and Drug Administration. Digital Health Center of Excellence and current guidance. 2026. <https://www.fda.gov/medical-devices/digital-health-center-excellence>
- [23] U.S. Food and Drug Administration, Office of Clinical Pharmacology. 2025 Annual Report. 2025. <https://www.fda.gov/media/191717/download>
- [24] U.S. Food and Drug Administration. Cybersecurity in Medical Devices: Quality Management System Considerations and Content of Premarket Submissions. 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-considerations-and-content-of-premarket-submissions>
- [25] International Organization for Standardization. ISO 13485: Medical devices - quality management systems. 2016. <https://www.iso.org/standard/59752.html>
- [26] International Organization for Standardization. ISO 14971: Medical devices - application of risk management. 2019. <https://www.iso.org/standard/72704.html>
- [27] International Electrotechnical Commission. IEC 60601-1: Medical electrical equipment - basic safety and essential performance. 2005+A1+A2. <https://webstore.iec.ch/publication/67497>
- [28] International Electrotechnical Commission. IEC 62366-1: Application of usability engineering to medical devices. 2015+A1. <https://webstore.iec.ch/publication/63111>
- [29] Collins GS, Moons KGM, Dhiman P, et al.. TRIPOD+AI statement. 2024. <https://www.tripod-statement.org/>
- [30] International Labour Organization. Safety and health at work in a changing climate and digital workplace. 2024-2026. <https://www.ilo.org/topics-and-sectors/safety-and-health-work>
- [31] World Health Organization. World report on vision and eye care resources. 2019/2026. <https://www.who.int/health-topics/blindness-and-vision-loss>
- [32] World Health Organization. Global Patient Safety Action Plan 2021-2030. 2021. <https://www.who.int/teams/integrated-health-services/patient-safety/policy/global-patient-safety-action-plan-2021-2030>
- [33] Damschroder LJ, Reardon CM, Widerquist MAO, Lowery J. The updated Consolidated Framework for Implementation Research. 2022. <https://doi.org/10.1186/s13012-022-01245-0>
- [34] Lopez-Otin C, Blasco MA, Partridge L, Serrano M, Kroemer G. Hallmarks of aging: an expanding universe. 2023. <https://doi.org/10.1016/j.cell.2022.11.001>
- [35] World Health Organization. Life course approach to health. 2025/2026. <https://www.who.int/our-work/life-course>

Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.